

LITTLE STREAM DETECTOR

LSD 3.0

Technical Documentation

Capabilities, operation and result interpretation

Property	Value
Release	3.0
Edition	User-focused final edition
Platform	Windows PowerShell 5.1 + .NET Framework
External multimedia tools	None required
Primary use	Native stream inspection and A/B comparison
Revision	19 September 2026

Purpose of this edition

This edition explains what LSD can analyze, how to operate the application, and how to interpret its reports, bitrate graphs and QP/quantizer distributions. Internal implementation details are intentionally limited to a short technical reference.

1. What LSD 3.0 Does

Little Stream Detector 3.0 is a native Windows media inspection application for examining video files without launching FFmpeg, FFprobe, MediaInfo or another external multimedia executable. It reads supported containers, identifies tracks, analyzes supported elementary streams and presents both human-readable and structured results.

Typical uses

Inspect a single file, compare an encode with a reference, verify stream structure after remuxing, examine bitrate behavior, and compare frame-level quantizer distributions.

Core capabilities

- Open a video from the file picker or by drag and drop.
- Analyze container, track, codec, timing, bitrate and frame-level information where supported.
- Keep the latest analysis as A Current and optionally compare it with an immutable B Reference.
- Display independent Summary A and Summary B reports, structured Streams data, JSON and a processing Log.
- Plot A/B bitrate profiles and show interval values in a mouse-hover tooltip.
- Display QP or codec-equivalent quantizer statistics and distributions in percentages or raw frame counts.
- Report unavailable information as N/A rather than inventing or estimating a value.
- Allow a running operation to be cancelled.

Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, VP8, VP9, MPEG-4 Part 2/XviD
Audio	AAC, MP3, PCM, AC-3, E-AC-3
Outputs	Summary A/B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Operating the Application

Basic workflow

1. Choose Open video or drop a supported media file into the drop area.
2. Wait for analysis to complete. The current result becomes A Current.
3. Review Summary A, Streams, JSON and Log as needed.
4. Inspect the bitrate profile and quantizer panel on the right.
5. To compare files, retain a result as B Reference and analyze the other file as A Current.
6. Use Remove Ref B when the reference is no longer needed.
7. Use Copy to copy the active report for use elsewhere.

Main controls

Control	Purpose
Open video	Selects a file and starts analysis.
Drop video here	Accepts a file by drag and drop.
Cancel	Stops an analysis that is currently running.
Copy	Copies the currently active report.
Remove Ref B	Clears the stored comparison reference.

A/B comparison model

- A Current is the most recently analyzed result.
- B Reference is a stored comparison result that remains unchanged while another file is analyzed.
- The interface supports one current result and one reference result at a time.
- Blue represents A Current. Gold represents B Reference.

Comparison tip

Use percentages when comparing files with different frame counts or durations. Use raw counts when investigating the exact number of frames at a particular quantizer value.

3. Understanding the Reports

View	What it provides
Summary A	Readable report for the current file.
Summary B	Readable report for the stored reference.
Streams	Structured information about native tracks.
JSON	Serializable metadata and stream information.
Log	Processing lifecycle, validation notes and first-failure messages.

Information commonly shown

- File name, modification time, size and duration.
- Detected container and native document type.
- Number and type of native tracks.
- Video dimensions, codec family and codec-specific configuration.
- Audio format, rates, channels and bitrate where available.
- Canonical analysis status and routing information.
- Frame types, keyframe information and codec-dependent quantizer statistics where analysis is complete.

How to use JSON and Log

JSON is useful when results need to be archived, compared or processed by another tool. Log is intended for diagnostics: use it when a file does not complete, a statistic is unavailable, or validation rejects part of the stream.

N/A is meaningful

An N/A value means LSD did not obtain a reliable value under the active parser and validation rules. It should not be interpreted as zero.

4. Bitrate Profile

The bitrate profile plots the amount of video data over presentation-time intervals. A Current is blue and B Reference is gold. This makes local peaks, quiet sections and differences between two files visible over the full duration.

Reading the graph

- Tall bars indicate intervals containing more encoded data.
- Low bars indicate intervals containing less encoded data.
- Peak bucket reports the largest interval value observed for the active file.
- The horizontal position corresponds to playback time.
- Hovering the graph displays the selected time interval and available A/B bitrate values.

What the graph can reveal

- Bitrate spikes around complex scenes or scene changes.
- Different bitrate allocation between an encode and its reference.
- Unexpectedly flat, sparse or irregular sections.
- Broad similarity after a lossless remux, allowing for documented timing or container differences.

Do not judge quality from bitrate alone

A higher instantaneous or average bitrate does not automatically mean higher visual quality. Codec family, encoder settings, source complexity and quantizer behavior must be considered together.

5. QP and Quantizer Distribution

The quantizer panel summarizes the codec-specific quantizer metric for analyzed frames. For AVC and HEVC the frame-level value is SliceQPY. Other supported codecs use their corresponding quantizer index.

Codec family	Histogram value
H.264 / AVC	Frame-level SliceQPY
H.265 / HEVC	Frame-level SliceQPY
AV1	Base Q Index
VP8	Base Q Index
VP9	Base Q Index
MPEG-4 Part 2	VOP quantizer

Statistics above the distribution

Statistic	Meaning
Frames	Number of frames contributing to the available quantizer analysis.
Avg	Arithmetic mean of the analyzed quantizer values.
Median	Middle quantizer value after sorting.
Mode	Most frequently occurring quantizer value.
Range	Lowest and highest recorded values.

Percentage and count modes

The fixed QP distribution header remains visible while the rows scroll. Click the header to switch the A/B columns between the two display modes:

Mode	Columns	Best use
Percentage	A % and B %	Compare the shape of distributions, especially when frame totals differ.
Count	A pcs and B pcs	See the exact number of frames at each quantizer value.

- The colored bars remain a relative visual representation of the distribution.
- A values are blue and B values are gold in both modes.
- The A and B columns remain aligned when the mode changes.

6. Interpreting Quantizer Results

Quantizer distributions are most useful as patterns, not as isolated numbers. A shift toward lower or higher values can be significant, but the meaning depends on the codec family, encoder behavior, content and other settings.

Practical interpretation

- A narrow distribution indicates that many frames were encoded within a relatively small quantizer range.
- A broad distribution indicates stronger variation in quantization across the file.
- A different mode shows that the most common frame quantizer changed.
- Different percentages with similar totals indicate a genuine change in distribution shape.
- Similar percentages with different raw counts can occur when the files contain different numbers of analyzed frames.
- For an unchanged elementary stream after a lossless remux, frame counts and quantizer histograms should normally remain equivalent.

Important comparison rule

A and B quantizer distributions are directly comparable only when both files use the same quantizer metric. Values from different codec families should not be treated as a one-to-one quality scale.

What QP does not prove by itself

- QP alone does not measure visible quality.
- QP alone does not establish encoder efficiency.
- The same numeric value can have different practical meaning across codecs or encoding structures.
- LSD does not decode pixels and does not calculate PSNR, SSIM or VMAF.

7. Codec and Audio Analysis

Video information

Family	Native analysis
H.264 / AVC	SPS/PPS/VUI, I/P/B and SliceQP
H.265 / HEVC	VPS/SPS/PPS, I/P/B and SliceQP
AV1	OBUs traversal, frame state, geometry and Base Q Index
VP8	Frame tag, KEY/INTER and Base Q Index
VP9	Superframes, frame state, references and Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/S, GMC and VOP quantizer

Audio information

Family	Validation and reporting
AAC	Configuration, core/output rate, channels, SBR, PS, payload and bitrate
MP3	Version/layer, bitrate, sample rate, padding and frame size
AC-3 / E-AC-3	Syncword, bitstream ID, rate, frame size, blocks and channels
PCM	Block alignment, sample format, duration and bitrate

8. Support Matrix and Limitations

Video	Matroska/WebM	MP4/MOV	fMP4	AVI
H.264 / AVC	Yes	Yes	Yes*	Yes, Annex B
H.265 / HEVC	Yes	Yes	Yes*	No
AV1	Yes	Yes	Yes*	No
VP8	Yes	No	No	No
VP9	Yes	Yes, vp09	Yes, vp09*	No
MPEG-4 Part 2	Yes	Yes	Yes*	Yes

* Fragmented MP4 reuses the existing codec analyzer when the sample entry and payload format are supported.

Known limitations

- Encrypted CENC samples and auxiliary encryption structures are not supported.
- Standalone media segments without an initialization moov box are not supported.
- Files that are still being written and live network streams are not supported.
- VP8 is supported only in Matroska/WebM.
- VP9 in AVI is not supported.
- AVI H.264/AVC support is intended for Annex B streams.
- AVI files containing AAC or E-AC-3 audio are not currently supported.
- Container-only color metadata is not used as a fallback.
- LSD does not decode picture pixels or calculate objective quality scores.

9. Validation, Troubleshooting and Good Practice

When an analysis is incomplete

8. Open Log and identify the first validation or parser failure.
9. Confirm that the file is complete and can be read from local storage.
10. Check whether the container/codecs combination is listed as supported.
11. Compare the number of analyzed frames with the expected duration and frame rate.
12. Treat disabled or N/A histograms as unavailable rather than as zero-valued results.

Useful expectations for a lossless remux

Check	Expected relationship
Video/audio sample count	Equal, or differences explained by container edits.
Payload bytes	Equivalent elementary-stream accounting.
Keyframe and I/P/B counts	Equal for unchanged elementary streams.
SliceQP/histogram	Equal for unchanged AVC/HEVC elementary streams.
Timestamps	Equivalent after documented edit-list effects.
Rejected samples/headers	Zero for a validated asset.

Best practice

Keep the source and comparison files clearly identified. Compare like with like, start with the same codec metric, and evaluate bitrate, frame counts and quantizer distribution together.

Appendix A. Concise Technical Reference

LSD uses PowerShell for orchestration and the WinForms interface, with embedded C# for native parsing and analysis. Supported containers are converted into a shared, bounded sample model before codec and audio analysis. Standard MP4 and fragmented MP4 use separate indexing routes but feed the same reporting, bitrate and A/B comparison layers.

Processing flow

File selection -> container detection -> sample indexing -> canonical media model -> codec/audio analysis -> validation -> reports and A/B graphs

Defensive behavior

- Sample ranges must remain inside the source file.
- Unavailable information is reported as N/A.
- Quantizer histograms remain unavailable when parser completeness checks fail.
- Fragmented MP4 support is limited to standard unencrypted files with initialization metadata and supported media fragments.

Terminology

Term	Definition
A Current	Most recently analyzed result.
B Reference	Stored immutable comparison result.
Canonical sample	Container-neutral bounded sample with timing and key state.
QP / quantizer distribution	Frequency distribution of the codec-specific quantizer metric.
pcs mode	Raw per-quantizer frame counts for A and B.
fMP4	Fragmented MP4 whose media samples are described by movie fragments.